



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Practical No. : 5

Aim –

Use JavaScript function types, scope, closures; apply try-catch; build a palindrome checker.

Software/Tools Required –

VSCode, Chrome, HTML5, CSS3, Javascript (ES6)

Experiment Program Code –

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Shopping Cart</title>
```

```
<style>
```

```
body {
```

```
font-family: Arial;
```

```
background: #f2f2f2;
```

```
}
```

```
.container {
```

```
width: 800px;
```

```
margin: 30px auto;
```

```
background: white;

padding: 20px;

border-radius: 10px;

box-shadow: 0 0 10px gray;

}
```

```
input {

padding: 8px;

width: 150px;

margin: 5px;

}
```

```
button {

padding: 10px;

cursor: pointer;

}
```

```
table {

width: 100%;

border-collapse: collapse;

margin-top: 20px;

}
```

```
table,th,td {

border: 1px solid black;
```

```
}
```

```
th {
```

```
background: #007bff;
```

```
color: white;
```

```
}
```

```
th,td {
```

```
padding: 10px;
```

```
text-align: center;
```

```
}
```

```
.result {
```

```
margin-top: 20px;
```

```
font-size: 18px;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2>Shopping Cart Calculator Sejal Lende 24070521035</h2>
```

```
<input type="text" id="name" placeholder="Product Name">
```

```
<input type="number" id="price" placeholder="Price">
```

```
<input type="number" id="qty" placeholder="Quantity">
```

```
<button onclick="addProduct()">Add Product</button>
```

```
<table id="cartTable">
```

```
  <tr>
```

```
    <th>ID</th>
```

```
    <th>Product</th>
```

```
    <th>Price</th>
```

```
    <th>Quantity</th>
```

```
    <th>Total</th>
```

```
  </tr>
```

```
</table>
```

```
<div class="result" id="result">
```

```
  <h3>Item Summary</h3>
```

```
  <ul id="summary"></ul>
```

```
  <h3>Expensive Products (Price > 1000)</h3>
```

```
  <ul id="expensive"></ul>
```

```
</div>
```

```
<script>
```

```
  let cart = [];
```

```
  function addProduct(){
```

```
    let name = document.getElementById("name").value;
```

```
    let price = parseFloat(document.getElementById("price").value);
```

```
    let qty = parseInt(document.getElementById("qty").value);
```

```
if(name=="" || isNaN(price) || isNaN(qty)) {  
    alert("Please enter all fields");  
    return;  
}
```

```
let product = {  
    id:cart.length+1,  
    name:name,  
    price:price,  
    quantity:qty  
};
```

```
cart.push(product);  
displayCart();  
document.getElementById("name").value="";  
document.getElementById("price").value="";  
document.getElementById("qty").value="";  
}
```

```
function displayCart() {  
    let table = document.getElementById("cartTable");  
    table.innerHTML = `  
        <tr>  
            <th>ID</th>  
            <th>Product</th>
```

```
<th>Price</th>
```

```
<th>Quantity</th>
```

```
<th>Total</th>
```

```
</tr>
```

```
`;
```

```
cart.forEach(function(item) {
```

```
  table.innerHTML+= `
```

```
    <tr>
```

```
      <td>${item.id}</td>
```

```
      <td>${item.name}</td>
```

```
      <td>${item.price}</td>
```

```
      <td>${item.quantity}</td>
```

```
      <td>${item.price*item.quantity}</td>
```

```
    </tr>
```

```
`;
```

```
});
```

```
let total = cart.reduce(function(sum, item) {
```

```
  return sum+(item.price*item.quantity);
```

```
},0);
```

```
let discount=0;
```

```
if(total >= 50000)
```

```
discount = total*0.20;
```

```
else if(total >= 20000)
```

```
discount = total*0.10;
```

```
else if(total >= 5000)
```

```
discount = total*0.05;
```

```
let finalAmount = total - discount;
```

```
document.getElementById("result").innerHTML =
```

```
,
```

```
<b>Total Amount :</b> ${total}<br>
```

```
<b>Discount :</b> ${discount.toFixed(2)}<br>
```

```
<b>Final Amount :</b> ${finalAmount.toFixed(2)}
```

```
`;
```

```
let summary = document.getElementById("summary");
```

```
summary.innerHTML="";
```

```
cart.map(function(item) {
```

```
    summary.innerHTML+=
```

```
    `- ${item.name} : ${item.price*item.quantity}</li>`;

```

```
});
```

```
let expensive = document.getElementById("expensive");
```

```
expensive.innerHTML="";
```

```
let exp = cart.filter(function(item) {  
  
    return item.price>1000;  
  
});
```

```
exp.forEach(function(item) {  
  
    expensive.innerHTML+=`<li>${item.name}</li>`;  
  
});  
  
}
```

```
</script>
```

```
</body>
```

```
</html>
```

Output –

Shopping Cart Calculator Sejal Lende 24070521035

ID	Product	Price	Quantity	Total
1	towel	99	5	495
2	handwash	90	12	1080
3	shirt	500	2	1000

Total Amount : 2575
Discount : 0.00
Final Amount : 2575.00

Case Study Title –

To study the functioning of Arrays and Loops in Java Script

a) Create and manipulate arrays: push, pop, shift, unshift, splice, slice. b) Use array methods: map, filter(), reduce(), forEach (.

Case Study: -

Create an array of numbers and find the maximum and minimum value.

Case Study Program Code –

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Domino's Style Billing Counter</title>
```

```
<style>
```

```
body {
```

```
    font-family: Arial, sans-serif;
```

```
    background-color: #f8f9fa;
```

```
    margin: 40px;
```

```
    color: #333;
```

```
}
```

```
h1 {
```

```
    color: #d32f2f;
```

```
    text-align: center;
```

```
}
```

```
.container {  
  
    background: #fff;  
  
    padding: 20px;  
  
    border-radius: 8px;  
  
    box-shadow: 0 2px 6px rgba(0,0,0,0.1);  
  
    max-width: 650px;  
  
    margin: auto;  
  
}
```

```
select, button {  
  
    padding: 10px;  
  
    margin: 5px 0;  
  
    border-radius: 4px;  
  
    border: 1px solid #ccc;  
  
    font-size: 14px;  
  
}
```

```
button {  
  
    background-color: #d32f2f;  
  
    color: white;  
  
    border: none;  
  
    cursor: pointer;  
  
}
```

```
button:hover {
```

```
background-color: #a00000;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 15px;
}

th, td {
    border: 1px solid #ddd;
    padding: 10px;
    text-align: center;
}

th {
    background-color: #f4f4f4;
}

.highlight {
    background-color: #ffe082;
    font-weight: bold;
}

.bill-summary {
    margin-top: 20px;
    padding: 15px;
```

```
background: #e3f2fd;

border-left: 4px solid #1976d2;

border-radius: 4px;

}

</style>

</head>

<body>

<div class="container">

  <h1>🍕 Domino's Style Billing Counter</h1>

  <label>Select Item:</label>

  <select id="menuSelect">

    <option value="Margherita Pizza|199">Margherita Pizza - ₹199</option>

    <option value="Farmhouse Pizza|349">Farmhouse Pizza - ₹349</option>

    <option value="Peppy Paneer Pizza|299">Peppy Paneer Pizza -
₹299</option>

    <option value="Veg Extravaganza Pizza|399">Veg Extravaganza Pizza -
₹399</option>

    <option value="Chicken Dominator Pizza|449">Chicken Dominator Pizza -
₹449</option>

    <option value="Cheese Burst Burger|149">Cheese Burst Burger -
₹149</option>
```

```
<option value="Veggie Burger|129">Veggie Burger - ₹129</option>
<option value="Chocolate Shake|179">Chocolate Shake - ₹179</option>
<option value="Strawberry Shake|159">Strawberry Shake - ₹159</option>
<option value="Cold Coffee|139">Cold Coffee - ₹139</option>
</select>
```

```
<button onclick="addItem()">Add Item</button>
<button onclick="generateBill()">Generate Bill</button>
```

```
<table id="itemsTable">
  <thead>
    <tr>
      <th>Item</th>
      <th>Price (₹)</th>
    </tr>
  </thead>
  <tbody></tbody>
</table>
```

```
<div id="billSummary" class="bill-summary"></div>
</div>
```

```
<script>
```

```
const items = [];
```

```
function addItem() {
```

```
    const selected = document.getElementById("menuSelect").value;
```

```
    const [name, price] = selected.split("|");
```

```
    items.push({ name, price: parseFloat(price) });
```

```
    updateTable();
```

```
}
```

```
function updateTable() {
```

```
    const tbody = document.querySelector("#itemsTable tbody");
```

```
    tbody.innerHTML = "";
```

```
    items.forEach(item => {
```

```
        const row =
```

```
`<tr><td>${item.name}</td><td>₹${item.price.toFixed(2)}</td></tr>`;
```

```
        tbody.innerHTML += row;
```

```
    });
```

```
}
```

```
function generateBill() {
```

```
    if (items.length === 0) {
```

```
    alert("No items added yet!");

    return;
}
```

```
const prices = items.map(item => item.price);

const maxPrice = Math.max(...prices);

const minPrice = Math.min(...prices);

const total = prices.reduce((sum, p) => sum + p, 0);
```

```
const maxItem = items.find(item => item.price === maxPrice);

const minItem = items.find(item => item.price === minPrice);
```

```
document.getElementById("billSummary").innerHTML = `

    <strong>Total Bill:</strong> ₹${total.toFixed(2)} <br>

        <strong>Most    Expensive    Item:</strong>    ${maxItem.name}

    (₹${maxItem.price.toFixed(2)}) <br>

        <strong>Least    Expensive    Item:</strong>    ${minItem.name}

    (₹${minItem.price.toFixed(2)})

`;
```

```
// Highlight max and min rows
```

```
const rows = document.querySelectorAll("#itemsTable tbody tr");
```

```
rows.forEach(row => {  
  
    const price = parseFloat(row.cells[1].innerText.replace("₹", ""));  
  
    row.classList.remove("highlight");  
  
    if (price === maxPrice || price === minPrice) {  
  
        row.classList.add("highlight");  
  
    }  
  
});  
  
}  
  
</script>  
  
</body>  
  
</html>
```

Output –



Domino's Style Billing Counter

Select Item: Strawberry Shake - ₹159



Add Item

Generate Bill

Item	Price (₹)
Margherita Pizza	₹199.00
Peppy Paneer Pizza	₹299.00
Chicken Dominator Pizza	₹449.00
Strawberry Shake	₹159.00

Total Bill: ₹1106.00

Most Expensive Item: Chicken Dominator Pizza (₹449.00)

Least Expensive Item: Strawberry Shake (₹159.00)

Conclusion –

The experiment was successfully performed to study JavaScript arrays, array manipulation methods, array iteration methods, loops and object handling. A Shopping Cart Calculator was developed to store product objects, calculate the total amount and apply discount logic. A Maximum and Minimum Finder was also implemented using array traversal to determine the largest and smallest values from an array of numbers.